

Due on tuesday, May 2nd, in class

----- ★ -----

1. Minkowski tensors.

- (a) [5 pts] Consider the Minkowski tensors

$$D^{\mu\nu}, \quad D^{\mu\nu} B_\sigma, \quad D^{\mu\sigma} B_\sigma, \quad D^{\mu\sigma} D_{\mu\sigma}, \quad C^\mu B_\mu, \quad C^\mu B_\sigma.$$

Explain the rank of each tensor.

Which of these, if any, are scalars?

- (b) [3 pts] Show how the tensor $T_\gamma^{\alpha\beta}$ transforms under the Lorentz transformation Λ .
- (c) [4 pts] A Minkowski tensor $U_{\mu\nu}$ is said to be symmetric if $U_{\mu\nu} = U_{\nu\mu}$. If a tensor is symmetric in one inertial frame, show that it is symmetric in all inertial frames.

2. Four-velocities.

- (a) [4 pts] Write down the four-velocity of a particle with nonzero mass m moving in the y direction with speed v . Show that this four-vector is timelike.
- (b) [5 pts] Find the four-velocity of a particle with nonzero mass m and velocity $\vec{u} = (c/2, c/2, c/2)$. Show that this four-vector is timelike. Is this particle massless?
- (c) [2 pts] What is the four-velocity of the particle with mass m in its own rest frame, i.e., in the frame where it is at rest?
- (d) [2 pts] Try writing down the four-velocity of a photon moving in the y direction. Is there a problem?

3. Collision/decay problems.

- (a) [6 pts] A neutral pion of rest mass m , while moving in the positive x direction, decays into two photons. One of the photons is emitted in the same direction as the original pion, and the other in the opposite direction. If the first photon has three times the energy of the second, find the pion's original speed.
- (b) [8 pts] Two photons have the same energy E_γ . They collide at an angle θ and create a single particle. Calculate the mass M of the final particle in terms of E_γ and θ .
- (c) [6 pts] Two balls of equal mass (m each) approach each other with equal but opposite velocities of magnitude v . Their collision is perfectly inelastic, so they stick together and form a single body of mass M . What is the velocity of the final body and what is its mass M ?
Find the mass M in the specific cases of $v = 0.01c$, $v = 0.5c$, and $v = 0.9c$.

4. [5 pts] Poincaré transformations.

If Λ is a 4×4 matrix representing a Lorentz transformation, then transformations of the type

$$x' = \Lambda x + a$$

are known as Poincaré transformations. Here x and x' are 4×1 column vectors (4-vectors) representing spacetime coordinates of events as seen from two different frames, and a is a 4×1 column vector. In other words, a Poincaré transformation is a combination of a Lorentz transformation plus a possible shift of the space and time coordinates. We will denote this transformation as (Λ, a) .

Show that the result of two Poincaré transformations (Λ_1, a_1) and (Λ_2, a_2) , applied successively, is the Poincaré transformation

$$(\Lambda_2 \Lambda_1, \Lambda_2 a_1 + a_2)$$